

Link Github: <https://github.com/Samnangalex06/semaphore.git>

Problem 1:

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snalex@snalex-GF63-Thin-10SC:~/Documents/I4/S1$ /usr/bin/python3 /home/snalex/Documents/I4/S1/05/problem1a.py
=== Problem 1: Entangled Particle Pairs Simulation

[P1] placed pair (P1-A, P1-B) size=2
[Consumer] took (P1-A, P1-B) size=0
[P3] placed pair (P3-A, P3-B) size=2
[P2] placed pair (P2-A, P2-B) size=4
[P1] placed pair (P1-A, P1-B) size=6
[P2] placed pair (P2-A, P2-B) size=8
shipped (P1-A, P1-B)

[Consumer] took (P3-A, P3-B) size=6
[P3] placed pair (P3-A, P3-B) size=8
[P2] placed pair (P2-A, P2-B) size=10
[P1] placed pair (P1-A, P1-B) size=12
[P1] placed pair (P1-A, P1-B) size=14
shipped (P3-A, P3-B)

[Consumer] took (P2-A, P2-B) size=12
[P3] placed pair (P3-A, P3-B) size=14
[P2] placed pair (P2-A, P2-B) size=16
[P1] placed pair (P1-A, P1-B) size=18
[P2] placed pair (P2-A, P2-B) size=20
[P3] placed pair (P3-A, P3-B) size=22
[P3] placed pair (P3-A, P3-B) size=24
shipped (P2-A, P2-B)

[Consumer] took (P1-A, P1-B) size=22
[P1] placed pair (P1-A, P1-B) size=24
[P3] placed pair (P3-A, P3-B) size=26
[P2] placed pair (P2-A, P2-B) size=28
[P1] placed pair (P1-A, P1-B) size=30
shipped (P1-A, P1-B)

[Consumer] took (P2-A, P2-B) size=28
[P2] placed pair (P2-A, P2-B) size=30
[P3] placed pair (P3-A, P3-B) size=32
[P1] placed pair (P1-A, P1-B) size=34
[P3] placed pair (P3-A, P3-B) size=36
^C
Stopped.
snalex@snalex-GF63-Thin-10SC:~/Documents/I4/S1$

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Problem 2:

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[P3] placed pair (P3-A, P3-B) size=36
^C
Stopped.
snalex@snalex-GF63-Thin-10SC:~/Documents/I4/S1$ /usr/bin/python3 /home/snalex/Documents/I4/S1/05/problem2a.py
H L O E H E O L O H L L L O E O H O O E L O H L O E O O H O L L E L O H L E L O O H L L O O L O E O O H L E H E O L L H L O E O O L
H E O H E L O H L O L E L L O L H L L O L E H L O L O E O O L L H O O E L O H E L H E O H L O L E L H L L O E L O H O L E H E O O H
L E O O L H O L O E H L L E O H L O L E L O H L E H O L E O H L L L O E O O L H E O L O H L L L E H O L O E L L L L L H O L E L O ^C
Exception ignored in: <module 'threading' from '/usr/lib/python3.12/threading.py'>
Traceback (most recent call last):
  File "/usr/lib/python3.12/threading.py", line 1622, in shutdown

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Problem 3:

